How to install ansible in AWS EC2:

Requirements: VPC with **2 subnets** (public and private **in different Availability Zone**) and **2 EC2s that runs ubuntu on it in the public subnet.**

Note:

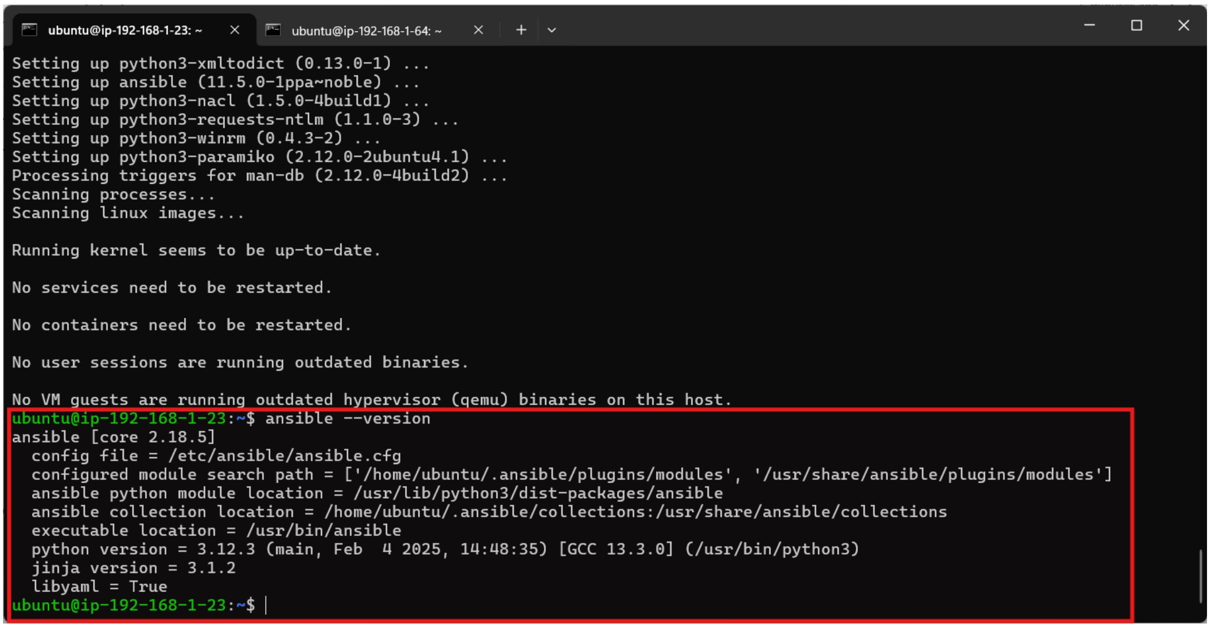
- **Master:** is referred to the machine that will deploy the ansible playbook (Control Node)
- **Slave:** is the **target machine** that Ansible configures (Managed/Client Node)

Step 1: update the EC2 by running the command "**Sudo apt update**"

Step 2: run these commands in the master machine to **install ansible:**

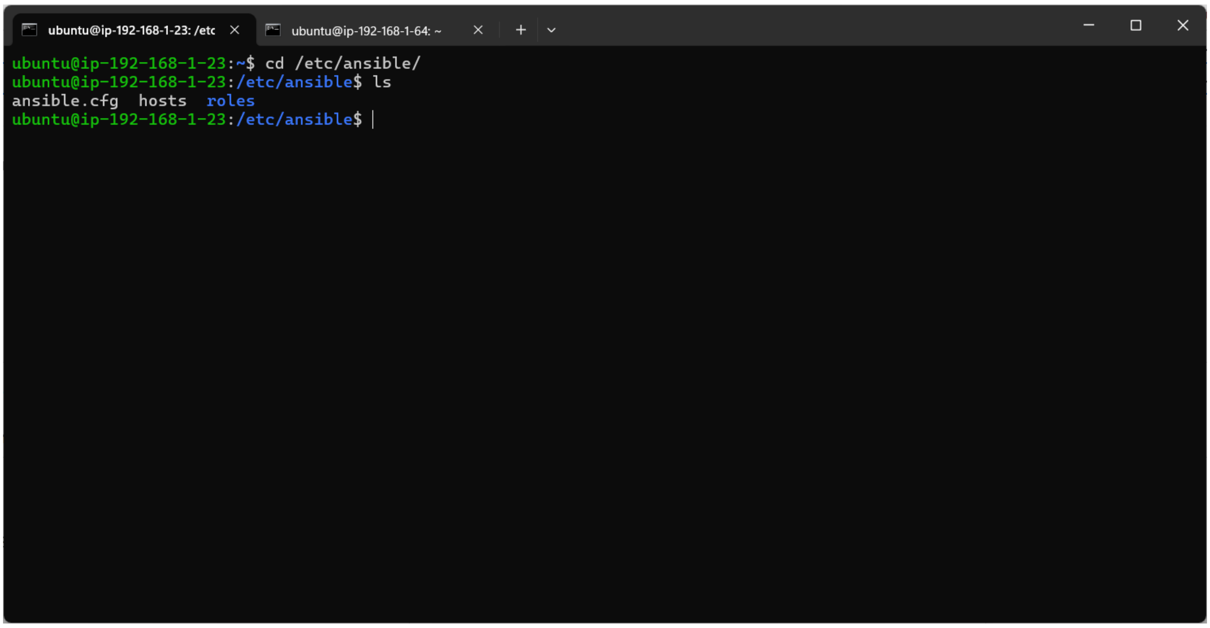
- sudo apt install software-properties-common
- sudo add-apt-repository --yes --update ppa:ansible/ansible
- sudo apt install ansible

step 3: to check that you install ansible right issue the command "**ansible --version**"



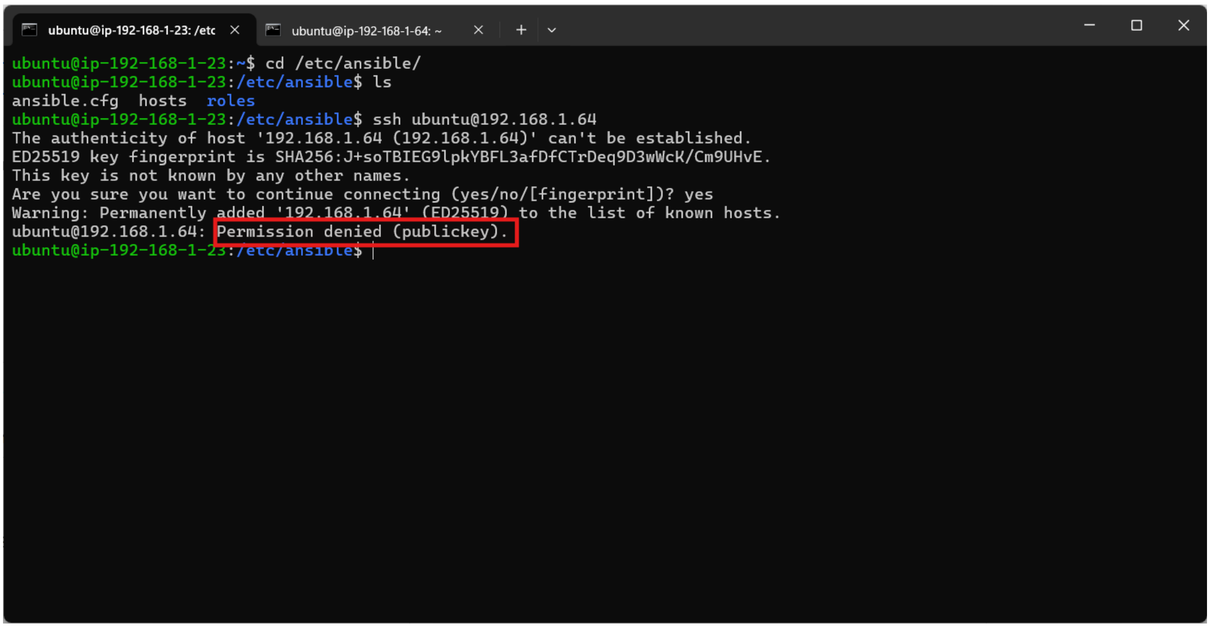
```
ubuntu@ip-192-168-1-23: ~        ubuntu@ip-192-168-1-64: ~
Setting up python3-xmltodict (0.13.0-1) ...
Setting up ansible (11.5.0-1ppa~noble) ...
Setting up python3-nacl (1.5.0-4build1) ...
Setting up python3-requests-ntlm (1.1.0-3) ...
Setting up python3-winrm (0.4.3-2) ...
Setting up python3-paramiko (2.12.0-2ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-192-168-1-23:~$ ansible --version
ansible [core 2.18.5]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Feb  4 2025, 14:48:35) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
ubuntu@ip-192-168-1-23:~$ |
```

Now you can check the ansible directory if you go to "**cd /etc/ansible**"

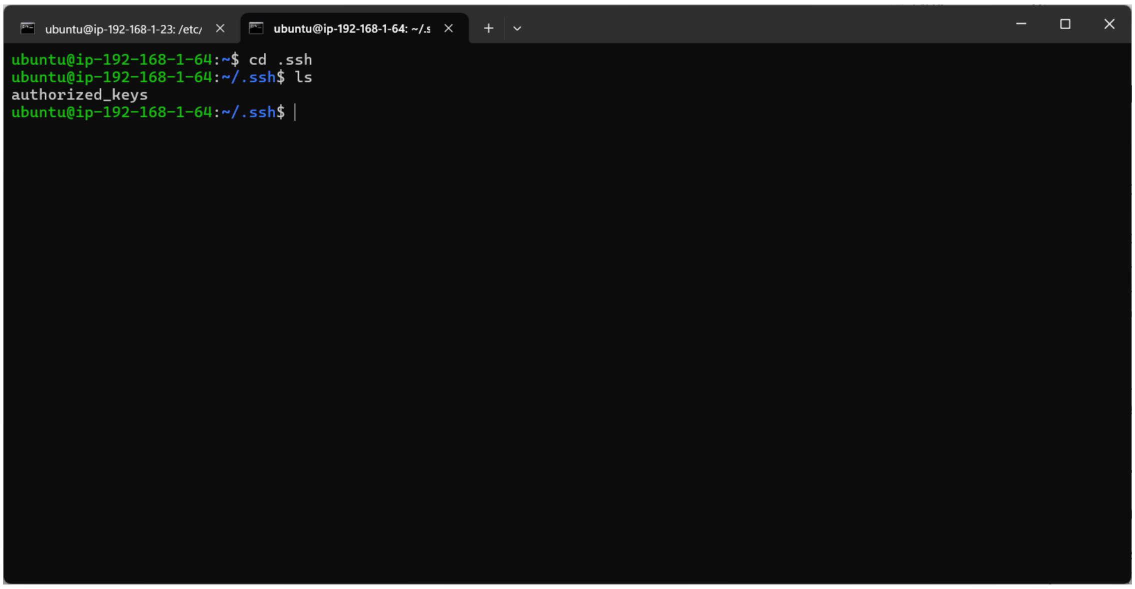


As you can see there are **3 files in the ansible directory**, but you will use two:

- **ansible.cfg** is the file that has the configuration for the ansible
- **hosts** where you list your Ip address for the servers or the machine that you want to remotely configure.



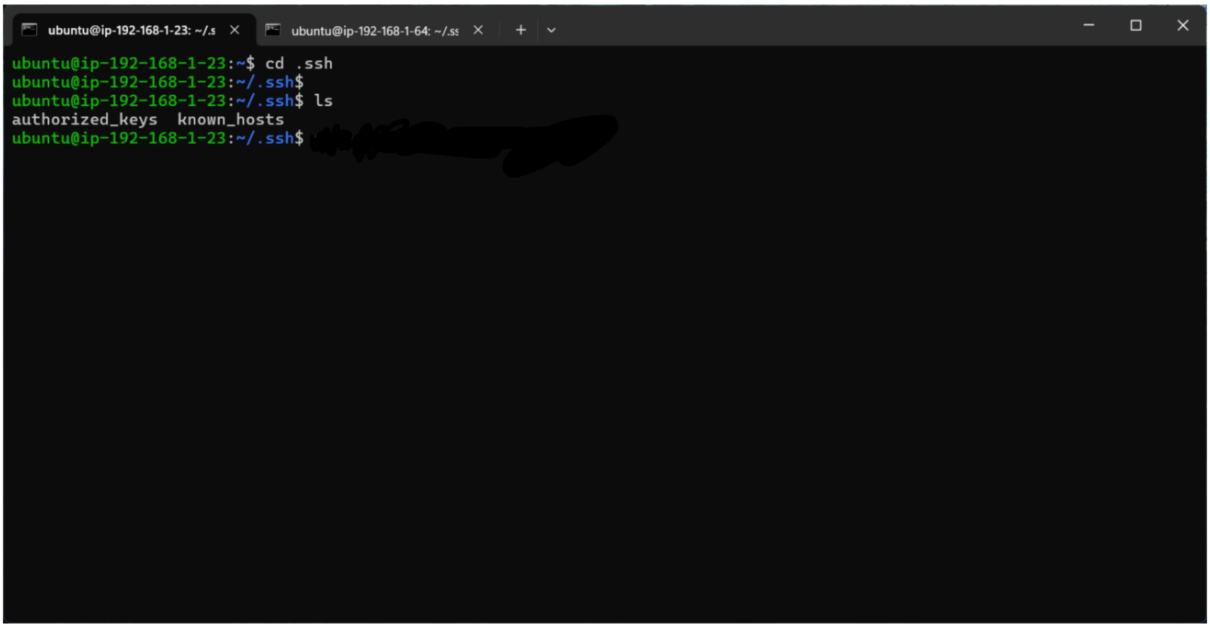
step 4: To access the slave machine via SSH, the slave must have the public key of the master machine.



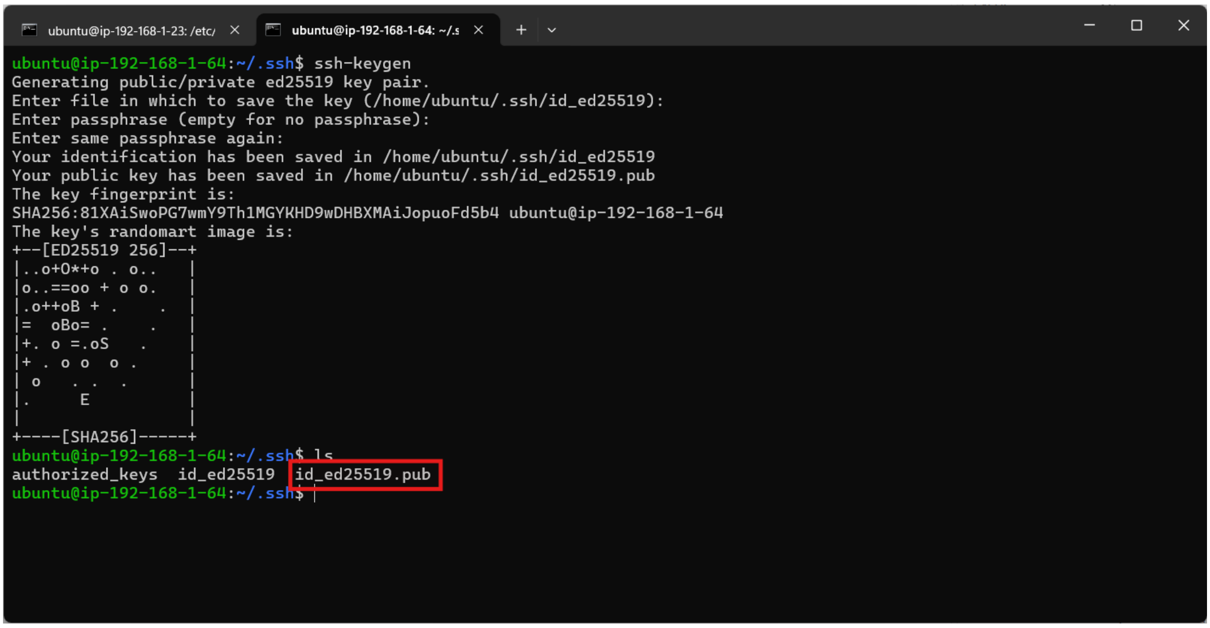
Step 5: you have to generate a rsa key from the master machine

First: go to the **master machine** in the home directory the go to .ssh file by issue the command “cd .ssh”

Note: if you don’t know were you are to go back to the home directory issue the command “cd ~” and you will be in the home directory.



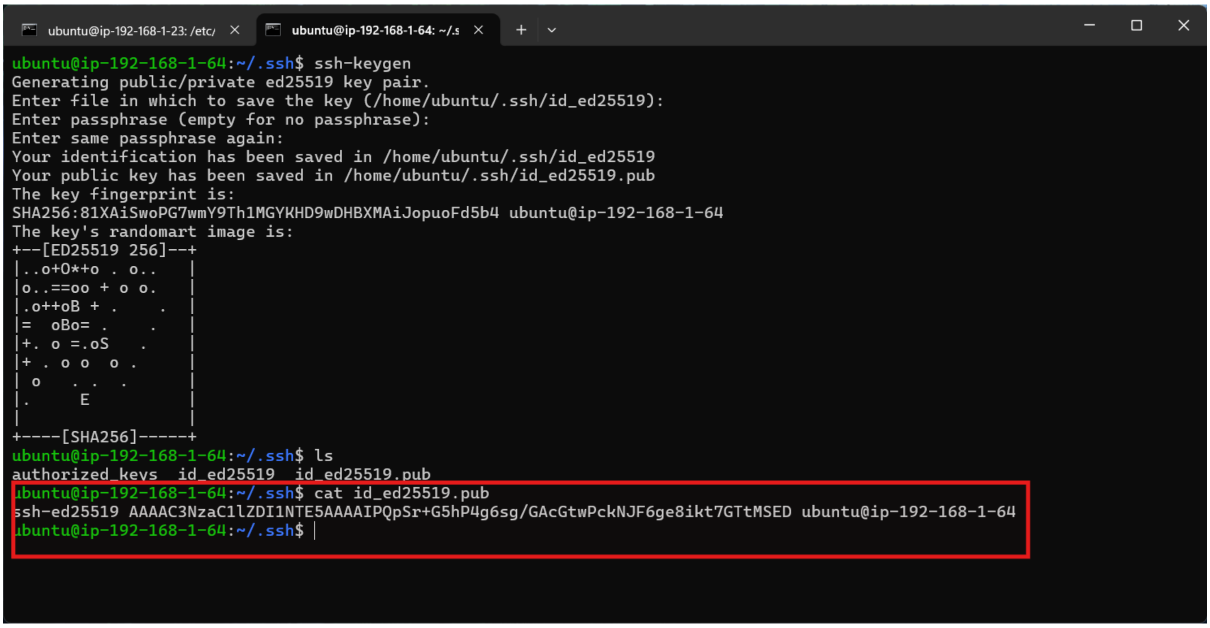
If you enter “**ls**” command, you will see that the .ssh folder has one file the authorized_keys.



```
ubuntu@ip-192-168-1-64:~/.ssh$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:81XAiSwoPG7wmY9Th1MGYKHD9wDHBXMAiJopuoFd5b4 ubuntu@ip-192-168-1-64
The key's randomart image is:
+--[ED25519 256]--+
|..o+O*+o . o..   |
|o..==oo + o o.   |
|.o++oB + .    .  |
|=  oBo= .     .  |
|+. o =.oS    .   |
|+ . o o  o .     |
| o   . .  .      |
|.     E          |
|                 |
+----[SHA256]-----+
ubuntu@ip-192-168-1-64:~/.ssh$ ls
authorized_keys  id_ed25519  id_ed25519.pub
ubuntu@ip-192-168-1-64:~/.ssh$ |
```

Second: run the command in the Master machine "**ssh-keygen**" then **press enter 3 time** to generate the key that you will use for ssh into the slave machine.

To check if you generate the key correctly enter "**ls**" you should have 3 files two of them like this "**id_.....**" and "**id_-----.pub**" you need the .pub key for the ssh to work.



```
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Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:81XAiSwoPG7wmY9Th1MGYKHD9wDHBXMAiJopuoFd5b4 ubuntu@ip-192-168-1-64
The key's randomart image is:
+--[ED25519 256]--+
|..o+O*+o . o..   |
|o..==oo + o o.   |
|.o++oB + .    .  |
|=  oBo= .     .  |
|+. o =.oS    .   |
|+ . o o  o .     |
| o   . .  .      |
|.     E          |
|                 |
+----[SHA256]-----+
ubuntu@ip-192-168-1-64:~/.ssh$ ls
authorized_keys  id_ed25519  id_ed25519.pub
ubuntu@ip-192-168-1-64:~/.ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPQpSr+G5hP4g6sg/GAcGtwPckNJF6ge8ikt7GTtMSED ubuntu@ip-192-168-1-64
ubuntu@ip-192-168-1-64:~/.ssh$ |
```
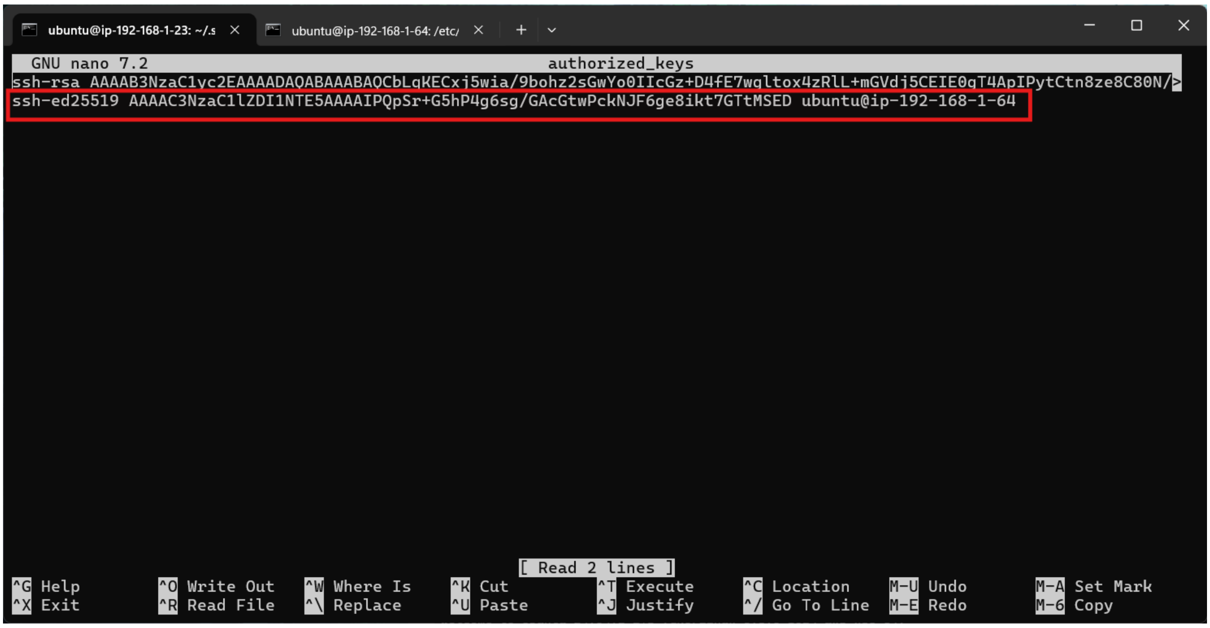
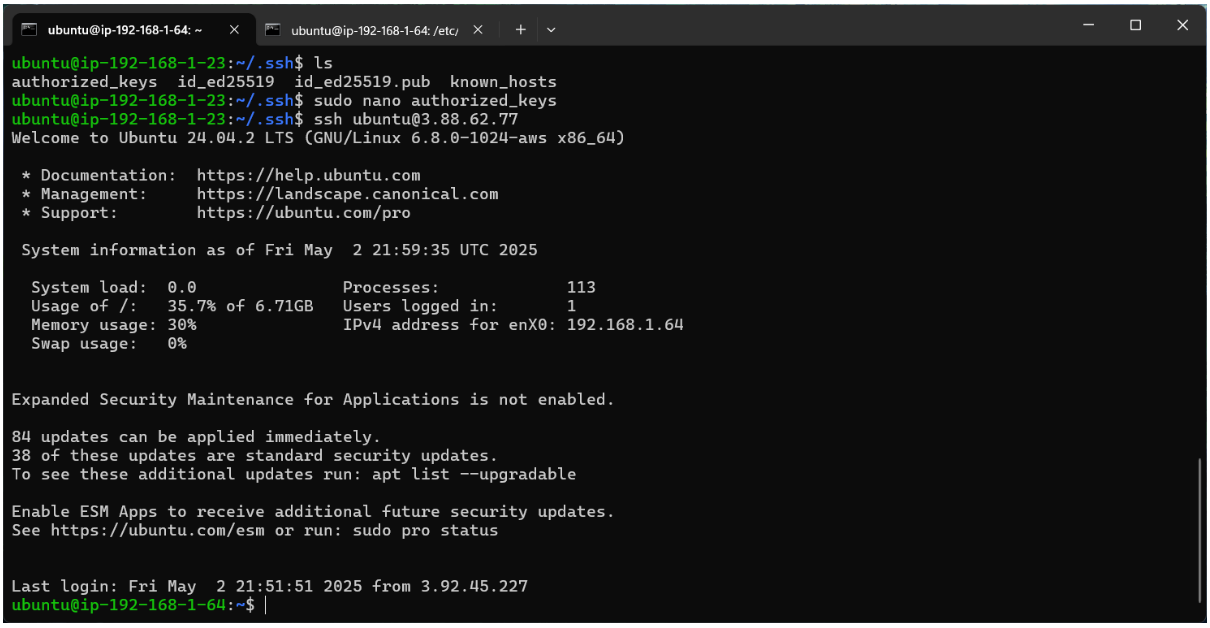

Third: copy the **.pub** by using the command "cat <id.pem>" and copy it.



Fourth: go to your **slave machine** and enter "**cd .ssh**" then "**ls**" to list what in the folder you will find the "**authorized_keys**" file open it using this command "**sudo nano authorized_keys**" then in the bottom **paste the key** you copy form the Master machine.

Note: to save and Exit click on "CTRL X then Y then press Enter".

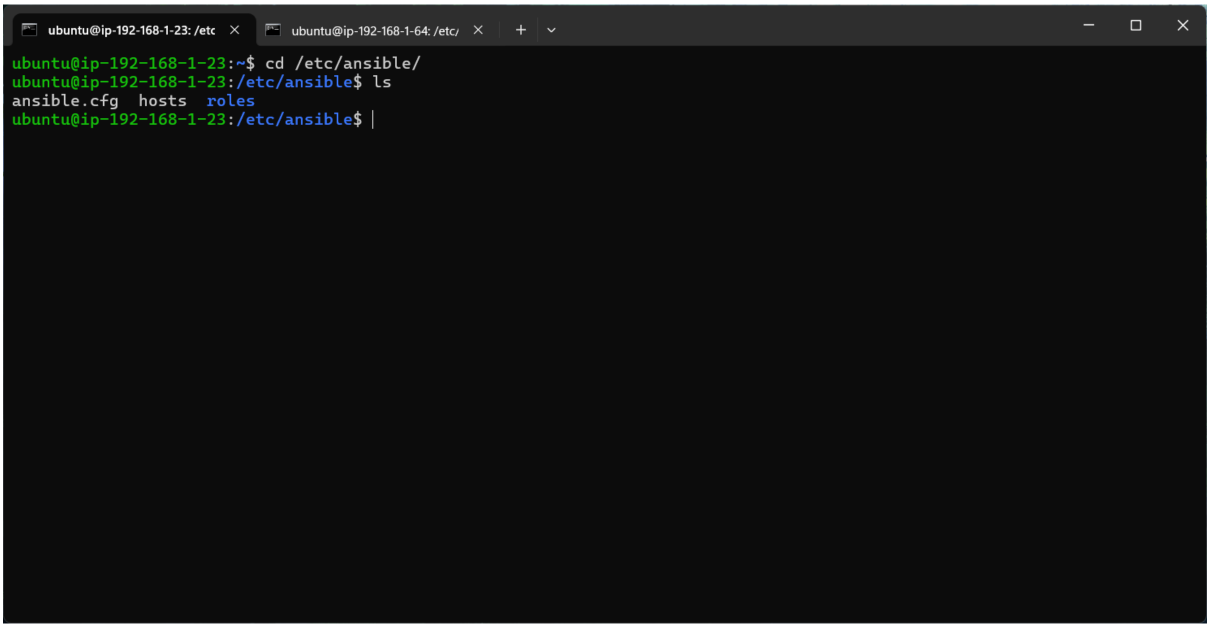


Now you can check that you done everything by ssh to the slave machine

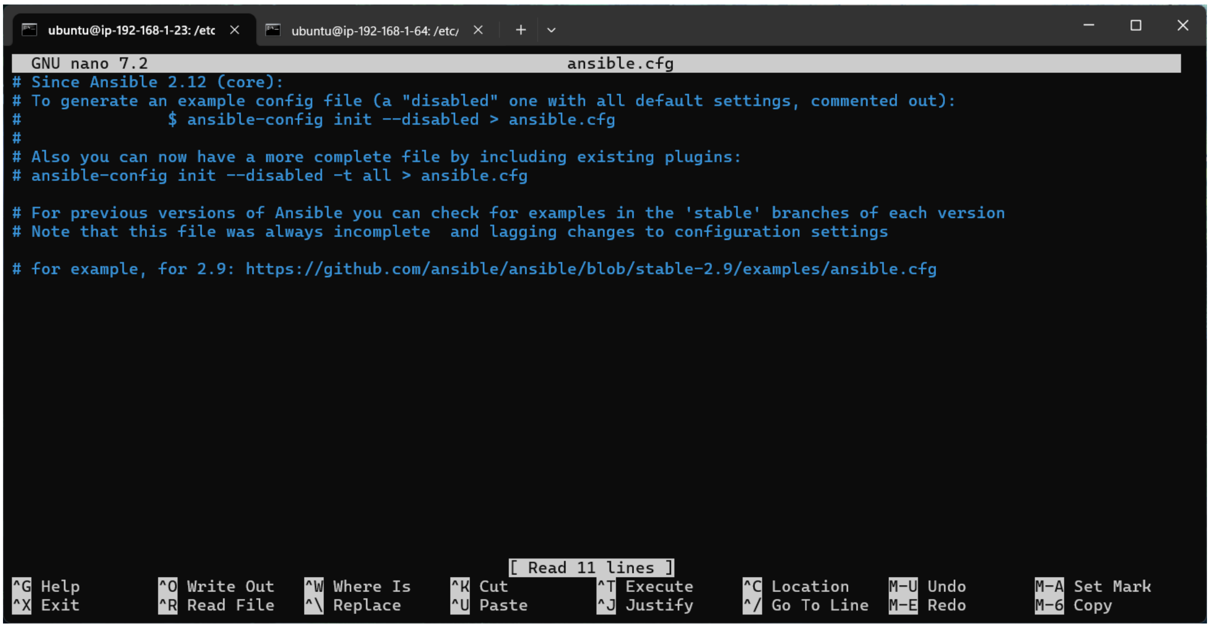
Enter the command “**ssh ubuntu@<the public ip for the slave machine>**”

As you can see, the ssh works.

Ansible Configuration:



Now we have to configure ansible go to the **master machine** in the ansible folder that located in the “**etc**” you can go there by issue the command “**cd /etc/ansible/**”



Step six: open the ansible.cfg by entering the command "sudo nano ansible.cfg" it will open like this

Copy the configuration and paste it in the file:

[defaults]

inventory = /etc/ansible/hosts

remote_user = **ubuntu** <u>***change it to the username of your machine**</u>

ask_pass = false

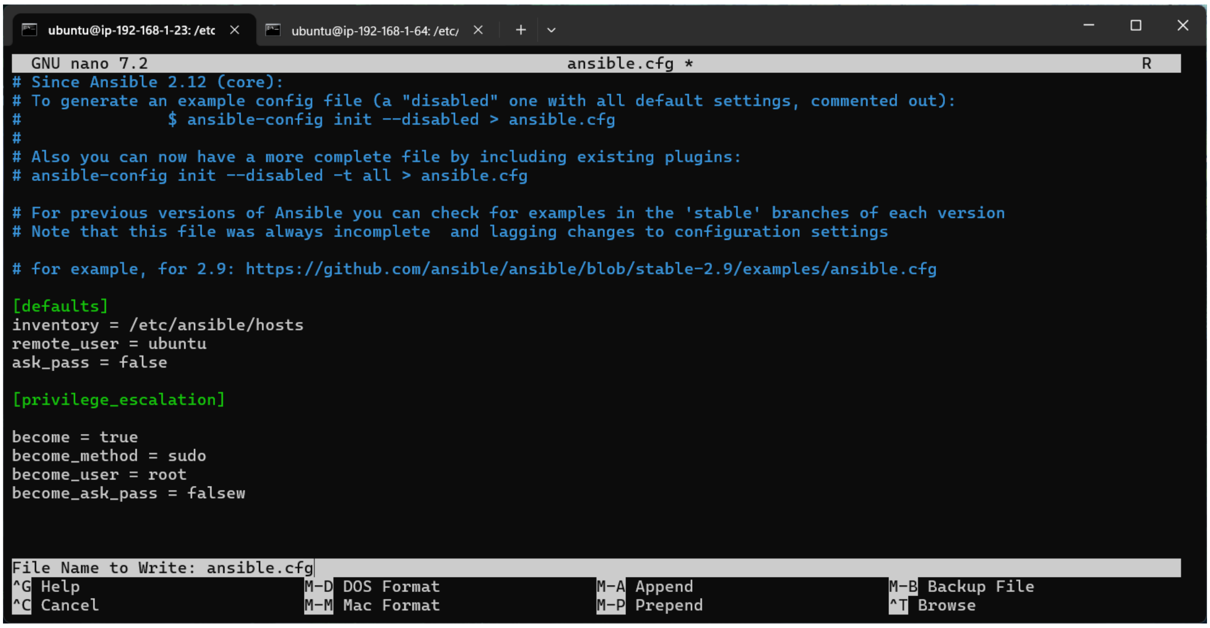
[privilege_escalation]

become = true

become_method = sudo

become_user = root

become_ask_pass = false



```
ubuntu@ip-192-168-1-23: /etc    ubuntu@ip-192-168-1-64: /etc/
 GNU nano 7.2                         ansible.cfg *                          R
# Since Ansible 2.12 (core):
# To generate an example config file (a "disabled" one with all default settings, commented out):
#               $ ansible-config init --disabled > ansible.cfg
#
# Also you can now have a more complete file by including existing plugins:
# ansible-config init --disabled -t all > ansible.cfg

# For previous versions of Ansible you can check for examples in the 'stable' branches of each version
# Note that this file was always incomplete  and lagging changes to configuration settings

# for example, for 2.9: https://github.com/ansible/ansible/blob/stable-2.9/examples/ansible.cfg

[defaults]
inventory = /etc/ansible/hosts
remote_user = ubuntu
ask_pass = false

[privilege_escalation]

become = true
become_method = sudo
become_user = root
become_ask_pass = falsew

File Name to Write: ansible.cfg
^G Help                         M-D DOS Format                  M-A Append                      M-B Backup File
^C Cancel                       M-M Mac Format                  M-P Prepend                     ^T Browse
```

Exit and save the changes.

```
ubuntu@ip-192-168-1-23:~$ cd /etc/ansible/
ubuntu@ip-192-168-1-23:/etc/ansible$ ls
ansible.cfg  hosts  roles
ubuntu@ip-192-168-1-23:/etc/ansible$ sudo nano ansible.cfg
ubuntu@ip-192-168-1-23:/etc/ansible$ cat ansible.cfg
# Since Ansible 2.12 (core):
# To generate an example config file (a "disabled" one with all default settings, commented out):
#               $ ansible-config init --disabled > ansible.cfg
#
# Also you can now have a more complete file by including existing plugins:
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[defaults]
inventory = /etc/ansible/hosts
remote_user = ubuntu
ask_pass = false

[privilege_escalation]

become = true
become_method = sudo
become_user = root
become_ask_pass = falsew
ubuntu@ip-192-168-1-23:/etc/ansible$
```
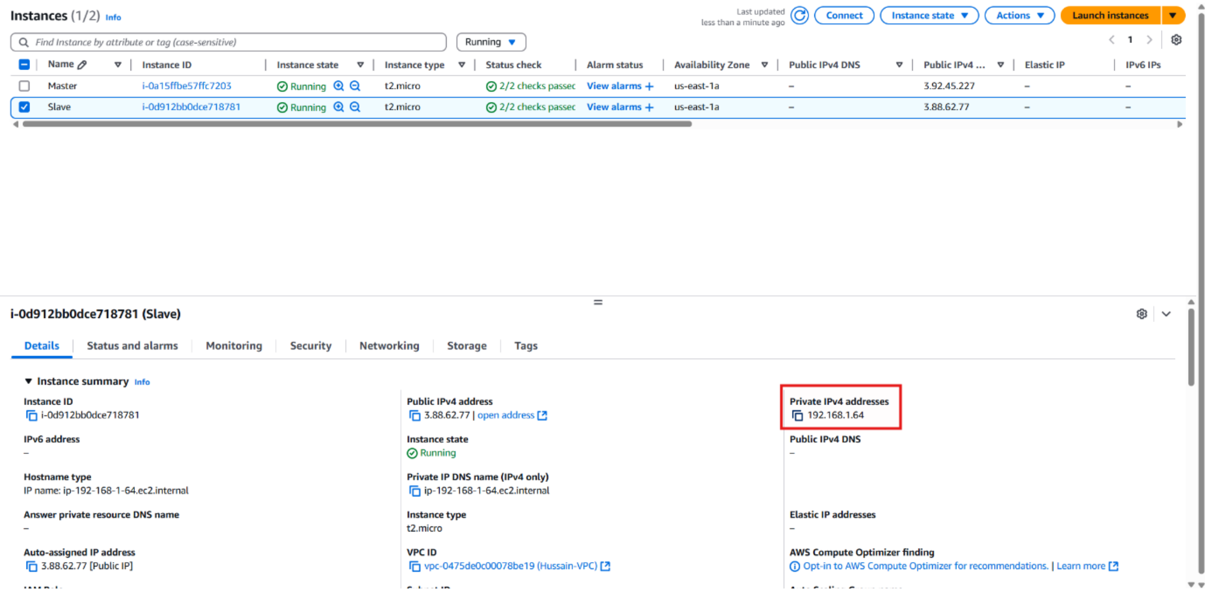
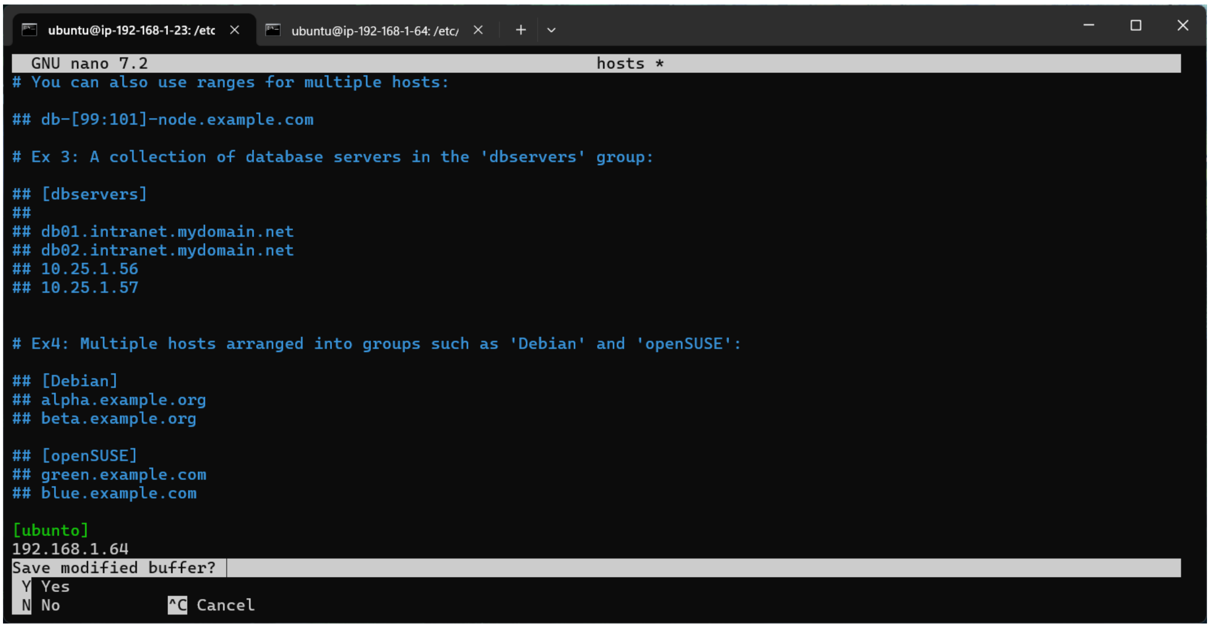

You can check if you save the changes by entering "**cat ansible.cfg**"

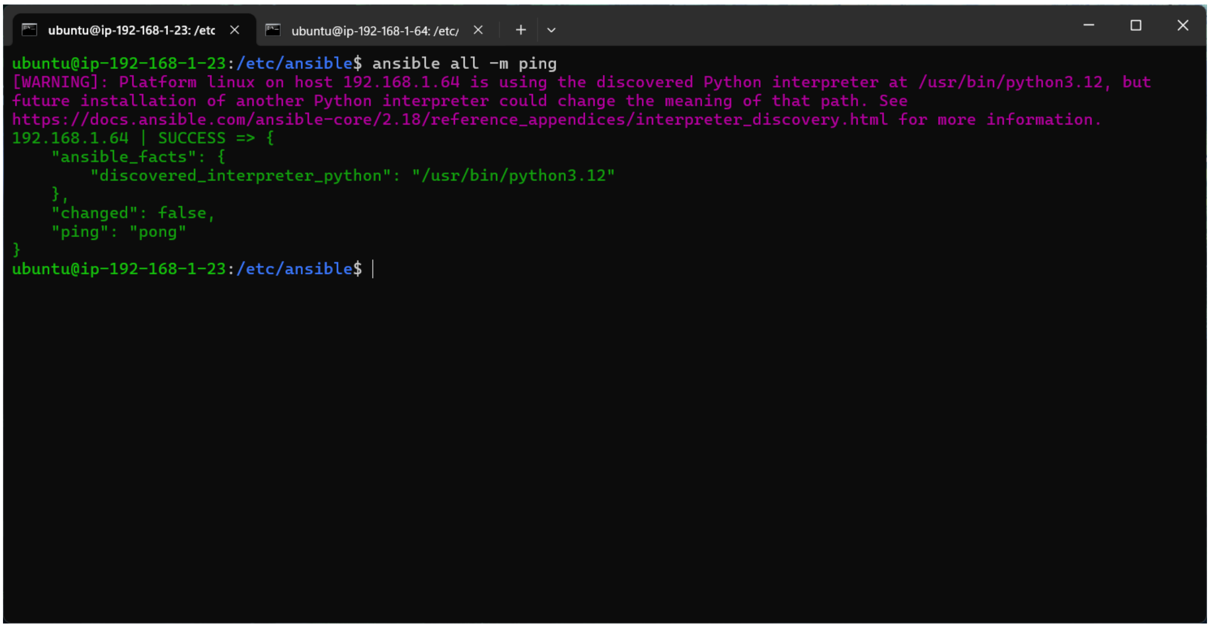
```
  GNU nano 7.2                                          hosts
# This is the default ansible 'hosts' file.
#
# It should live in /etc/ansible/hosts
#
#   - Comments begin with the '#' character
#   - Blank lines are ignored
#   - Groups of hosts are delimited by [header] elements
#   - You can enter hostnames or ip addresses
#   - A hostname/ip can be a member of multiple groups

# Ex 1: Ungrouped hosts, specify before any group headers:

## green.example.com
## blue.example.com
## 192.168.100.1
## 192.168.100.10

# Ex 2: A collection of hosts belonging to the 'webservers' group:

## [webservers]
## alpha.example.org
## beta.example.org
## 192.168.1.100
## 192.168.1.110

# If you have multiple hosts following a pattern, you can specify
                                            [ Read 54 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy
```

Step eight: open the hosts file by suing "sudo nano hosts" then copy the private ip for the slave machine.





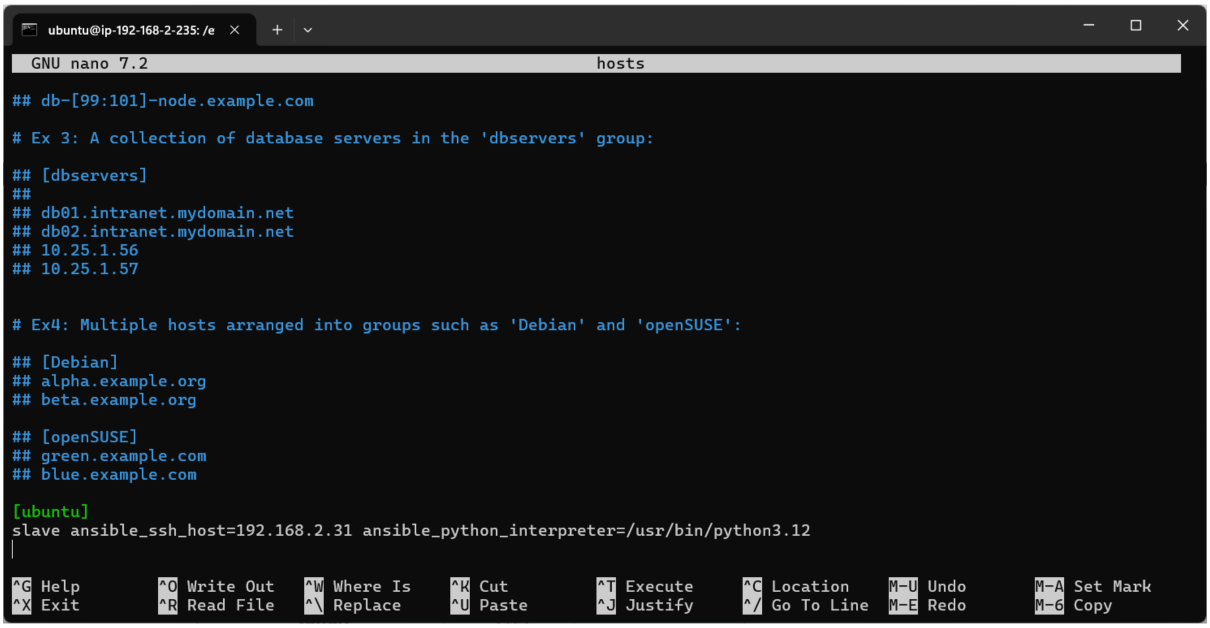
in the bottom of the file enter a name for the ip that you want to manage like [ubuntu] . Then under it enter the ip of the slave machine you copy it then exit and save the changes.



Step nine: to check if you done everything correctly you will ping the slave machine

By entering the command “ansible all -m ping”.

- ✓ If you have done everything right, it will give you SUCCESS massage.



To remove the warning and give it a name for the client machine in the hosts file.